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7.1 INTRODUCTION

DC machines are electro mechanical energy conversion devices which can operate as generators converting mechanical energy given to them from a prime mover to electrical energy. They can also operate as DC motors, taking electrical energy from DC supply and converting it into mechanical energy to drive a mechanical load. Nowadays DC motors are widely used because of their simpler and flexible drive systems such as electric traction, cranes, etc. The extension of these DC machines leads to the development of brushless DC machines which are mostly used in modern scenario.

As said above, an electrical machine which converts mechanical energy into electrical energy is called an electrical generator, while the electrical machine which converts electrical energy into mechanical energy is called an electrical motor.

The electrical machines which work on AC supply are AC machines whereas the electrical energy of direct type are called DC machines. In this chapter we are going to study about DC machines.

DC machines are classified into two types:

1) DC generator
2) DC motor

First, we will study the basic principle of DC generators.

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